

Memory Management

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Notes

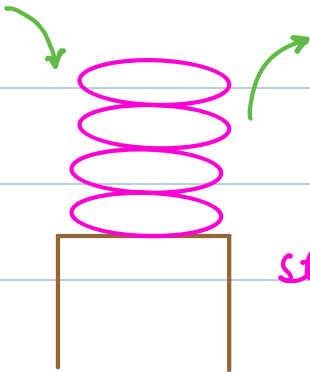
"Our greatest weakness lies in giving up. The most certain way to succeed is always to try just one more time."

Thomas Edison

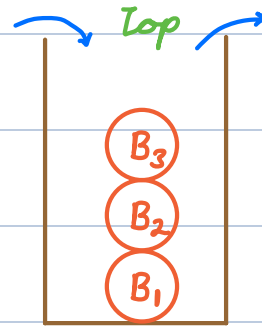
Forbes



Introduction to Stack



Stack of Plates



Stack of
chairs

Last in first out $\rightarrow$ LIFO

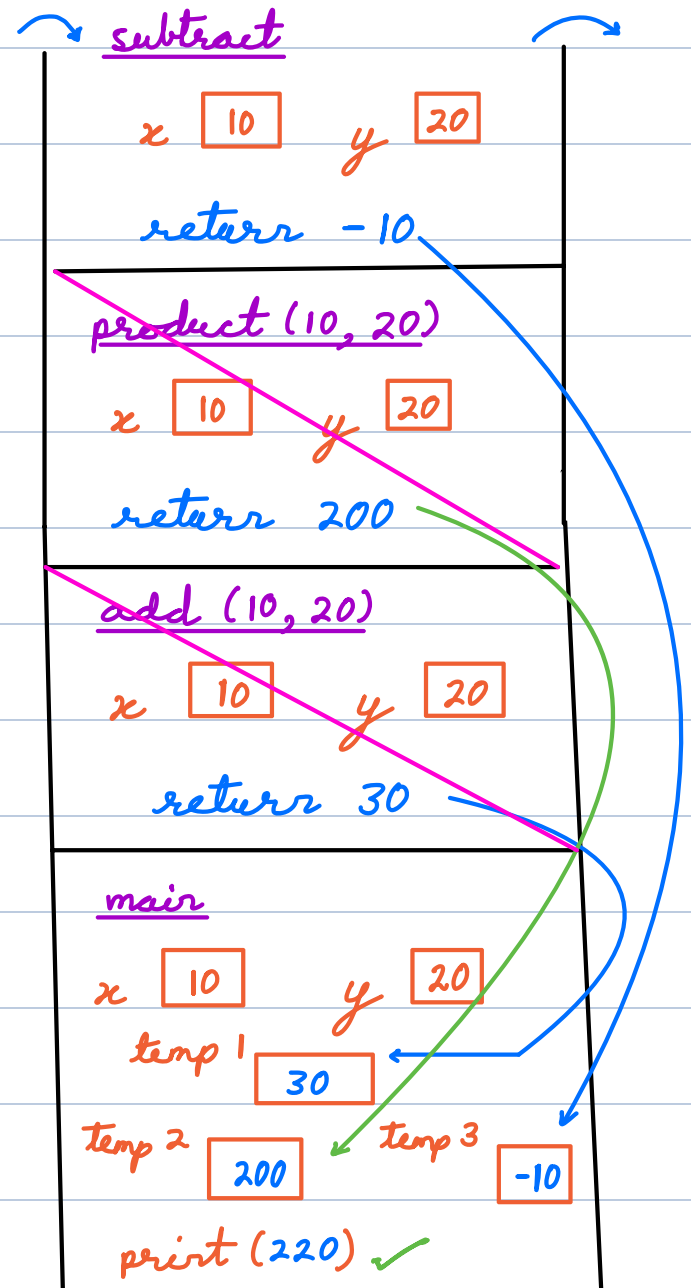
Only have access to
top element.



Introduction to Call Stack

```
int add(int x, int y) {  
    return x + y;  
}  
  
int product(int x, int y) {  
    return x * y;  
}  
  
int subtract(int x, int y) {  
    return x - y;  
}  
  
public static void main() {  
    int x = 10;  
    int y = 20;  
    int temp1 = add(x, y);  
    int temp2 = product(x, y);  
    int temp3 = subtract(x, y);  
    System.out.println(temp1 + temp2 + temp3);  
}
```

```
def add(x, y):  
    return x + y  
  
def product(x, y):  
    return x * y  
  
def subtract(x, y):  
    return x - y  
  
def main():  
    x = 10  
    y = 20  
    temp1 = add(x, y)  
    temp2 = product(x, y)  
    temp3 = subtract(x, y)  
    print(temp1 + temp2 + temp3)  
  
if __name__ == "__main__":  
    main()
```



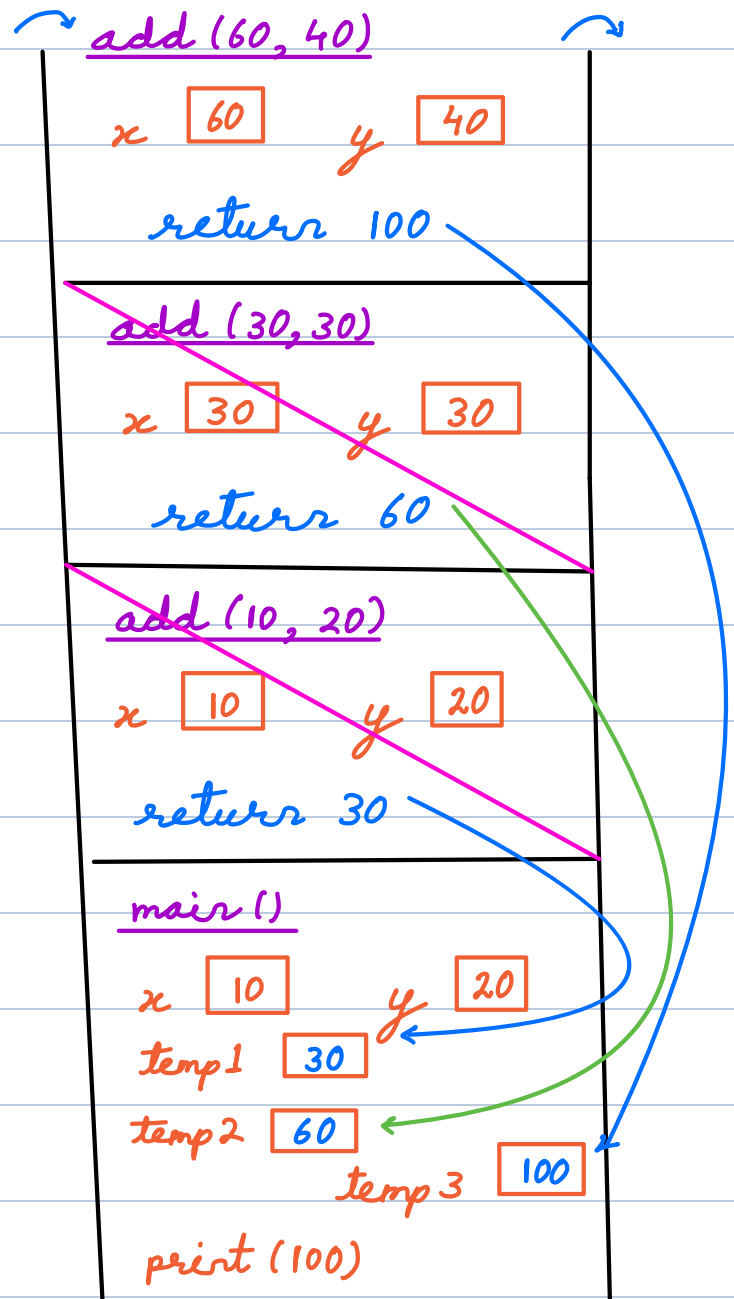


Java:

```
int add(int x, int y) {  
    return x + y;  
}  
  
public static void main() {  
    int x = 10;  
    int y = 20;  
    int temp1 = add(x, y);  
    int temp2 = add(temp1, 30);  
    int temp3 = add(temp2, 40);  
    System.out.println(temp3);  
}
```

Python:

```
def add(x, y):  
    return x + y  
  
def main():  
    x = 10  
    y = 20  
    temp1 = add(x, y)  
    temp2 = add(temp1, 30)  
    temp3 = add(temp2, 40)  
    print(temp3)  
  
if __name__ == "__main__":  
    main()
```





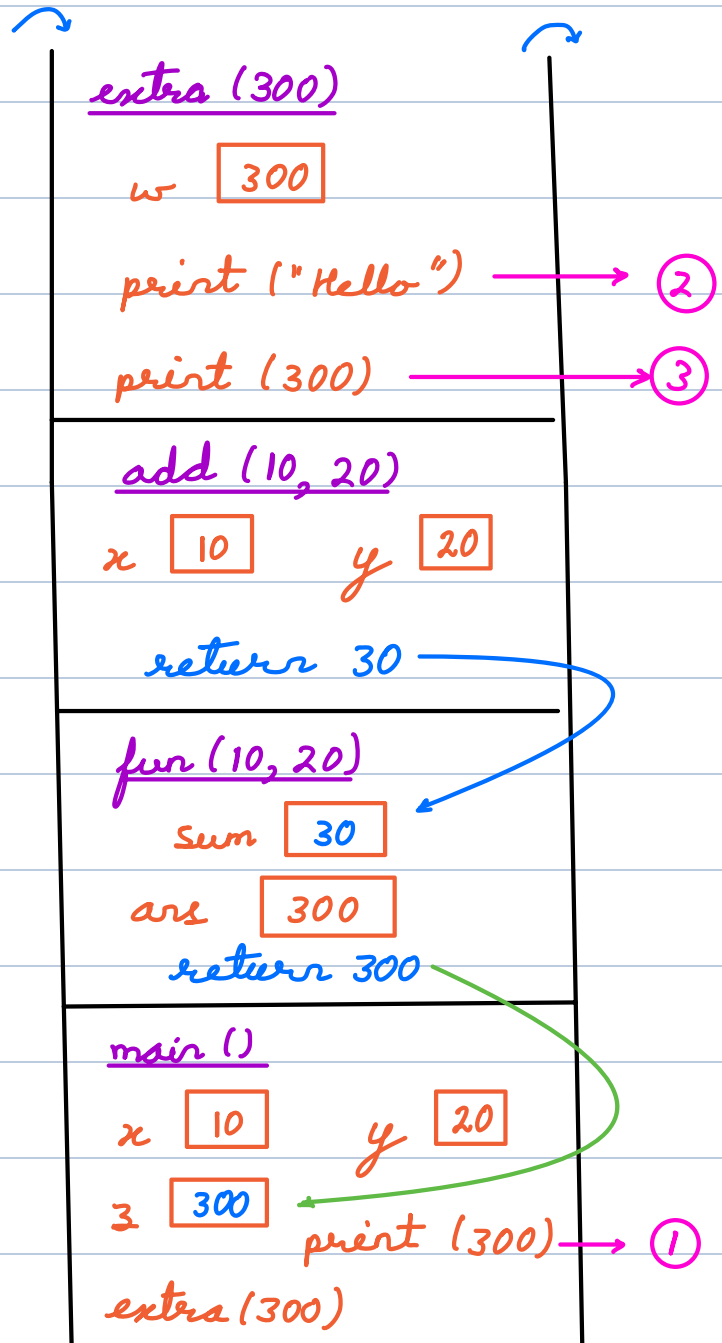
Java:

static

```
int add(int x, int y) {  
    return x + y;  
}  
  
static int fun(int a, int b) {  
    int sum = add(a, b);  
    int ans = sum * 10;  
    return ans;  
}  
  
static void extra(int w){  
    System.out.println("Hello");  
    System.out.println(w);  
}  
  
public static void main() {  
    int x = 10;  
    int y = 20;  
    int z = fun(x, y);  
    System.out.println(z);  
    extra(z);  
}
```

Python:

```
def add(x, y):  
    return x + y  
  
def fun(a, b):  
    sum = add(a, b)  
    ans = sum * 10  
    return ans  
  
def extra(w):  
    print("Hello")  
    print(w)  
  
def main():  
    x = 10  
    y = 20  
    z = fun(x, y)  
    print(z)  
    extra(z)  
  
if __name__ == "__main__":  
    main()
```





What is a reference variable ?

Imagine you have a piece of paper with instructions to find a particular book in a library. This piece of paper isn't the book itself; it just tells you where the book is located. A reference variable works in a similar way in programming. Instead of holding actual data directly (like numbers or text), a reference variable holds the address or the location of the data stored in the memory of the computer.

Just like how you'd follow the instructions on the paper to find your book, the computer uses the address stored in the reference variable to locate and access the actual data.

Think of it like giving someone a map to a treasure instead of carrying the treasure around yourself. When programs use reference variables, they're using "maps" that point to where the data is stored, rather than moving the actual data. This makes handling big or complex information, like lists, much faster and easier, just as it's easier to hand someone a map than to move the treasure every time someone wants to see it.

Types of Memory

1. Stack -

All the primitive data type and reference variables will be stored in stack.

2. Heap -

The data pointed by that reference variable is stored in heap like, arrays, arrayList, etc are created inside heap. (heap stores complex data)



Java:

```
public static void main() {  
    int x = 10;  
    int[] ar = new int[3];  
    System.out.println(ar); // #ad1  
    System.out.println(ar[2]); // 0  
    ar[1] = 7; ←  
}
```

Stackmain()

x 10

ar #ad

Heap

0	0	0
---	--------------	---

7

Python

```
def main():  
    x = 10  
    ar = [0] * 3 # Creating a list of size 3 initialized with zeros  
    print(ar)    # #ad1  
    print(ar[2]) # 0  
    ar[1] = 7  
  
if __name__ == "__main__":  
    main()
```

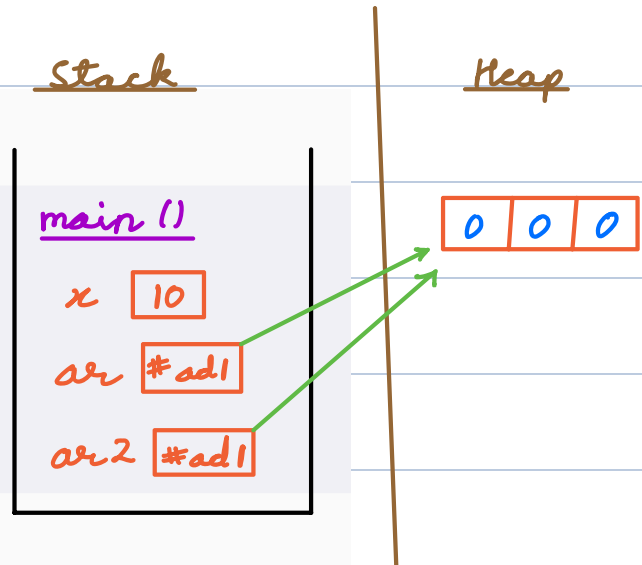


Java:

```
public static void main() {  
    int x = 10;  
    int[] ar = new int[3];  
    int[] ar2 = ar;  
    System.out.println(ar); // 4k  
    System.out.println(ar2); // 4k  
}
```

Python:

```
def main():  
    x = 10  
    ar = [0] * 3 # Creating a list of size 3 initialized with zeros  
    ar2 = ar     # Assigning ar to ar2  
    print(ar)  
    print(ar2)  
  
if __name__ == "__main__":  
    main()
```



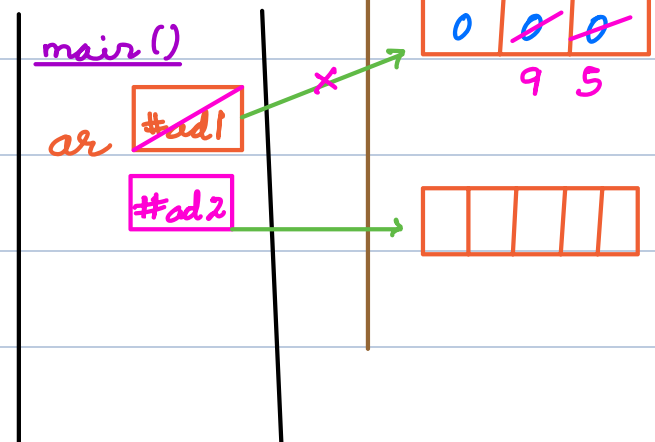


Java:

```
public static void main() {  
    int[] ar = new int[3];  
    System.out.println(ar); // 4k (#ad1)  
    ar[1] = 9;  
    ar[2] = 5;  
  
    ar = new int[5];  
    System.out.println(ar); // 7k (#ad2)  
}
```

Python:

```
def main():  
    ar = [0] * 3  
    print(ar)  
    ar[1] = 9  
    ar[2] = 5  
  
    ar = [0] * 5  
    print(ar)  
  
if __name__ == "__main__":  
    main()
```

StackHeap

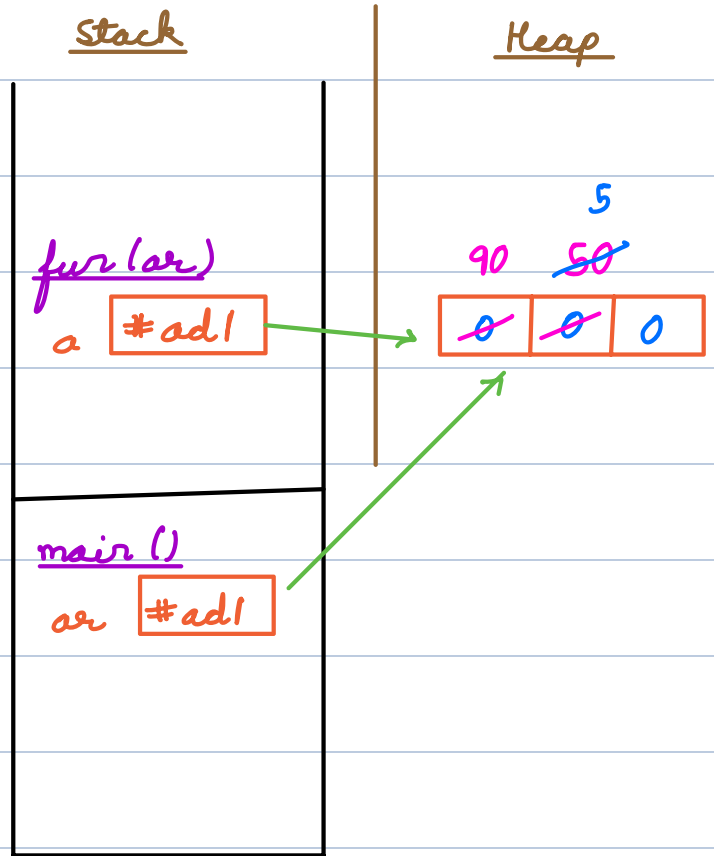


Java:

```
static void fun(int[] a){  
    System.out.println(a); // 9k (#ad1)  
    a[1] = 5;  
}  
  
public static void main() {  
    int[] ar = new int[3];  
    System.out.println(ar); // 9k (#ad1)  
    ar[0] = 90;  
    ar[1] = 50;  
    fun(ar);  
    System.out.println(ar[1]); // 5  
}
```

Python:

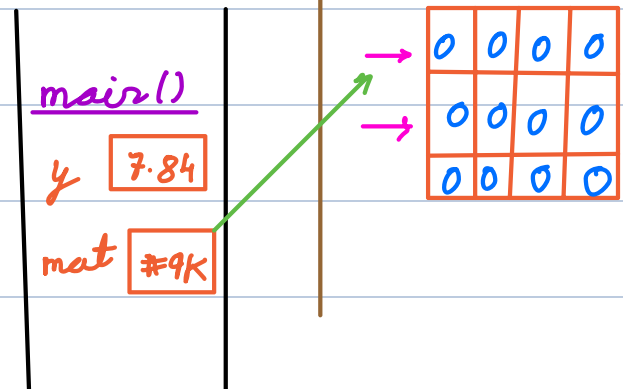
```
def fun(a):  
    print(a)  
    a[1] = 5  
  
def main():  
    ar = [0] * 3  
    print(ar)  
    ar[0] = 90  
    ar[1] = 50  
    fun(ar)  
    print(ar[1])  
  
if __name__ == "__main__":  
    main()
```





Java:

```
public static void main() {  
    float y = 7.84f;  
    int[][] mat = new int[3][4];  
    System.out.println(mat); // 9k ✓  
    System.out.println(mat[1]); // 3k  
    System.out.println(mat[1][3]); // 0 ✓  
}
```



Python:

```
def main():  
    y = 7.84  
    mat = [[0]*4 for _ in range(3)]  
    print(mat)  
    print(mat[1])  
    print(mat[1][3])  
  
if __name__ == "__main__":  
    main()
```



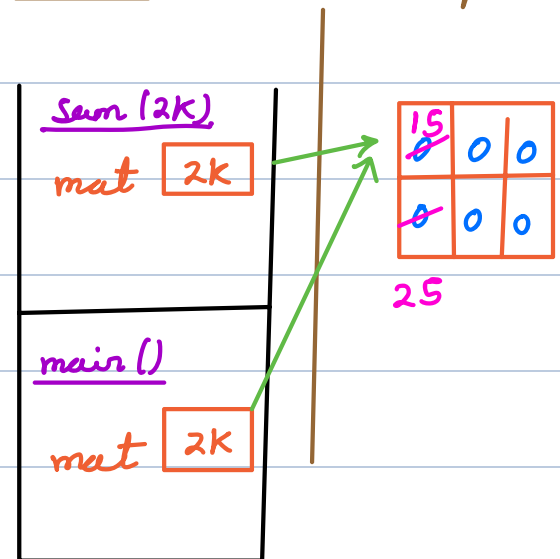
Java:

```
static void sum(int[][] mat){  
    System.out.println(mat); // 2k  
    System.out.println(mat[0][0] + mat[1][0]); // 40  
}  
  
public static void main() {  
    int[][] mat = new int[2][3];  
    mat[0][0] = 15;  
    mat[1][0] = 25;  
    sum(mat);  
}
```

$15 + 25 = 40$

Python:

```
def sum(mat):  
    print(mat)  
    print(mat[0][0] + mat[1][0])  
  
def main():  
    mat = [[0] * 3 for _ in range(2)]  
    mat[0][0] = 15  
    mat[1][0] = 25  
    sum(mat)  
  
if __name__ == "__main__":  
    main()
```

StackHeap



Java:

```
static int sumOfRow(int[] arr){
    System.out.println(arr); // 7k
    int sum = 0;
    for (int i = 0; i < arr.length; i++){
        sum = sum + arr[i];
    }
    return sum;
}

public static void main() {
    int[][] mat = new int[2][3];
    mat[0][0] = 9;
    mat[0][1] = 5;
    mat[0][2] = 1;
    int ans = sumOfRow(mat[0]); // 7k
    System.out.println(ans); // 15
}
```

Python:

```
def sum_of_row(arr):
    print(arr)
    total = 0
    for num in arr:
        total += num
    return total

def main():
    mat = [[0] * 3 for _ in range(2)]
    mat[0][0] = 9
    mat[0][1] = 5
    mat[0][2] = 1
    ans = sum_of_row(mat[0])
    print(ans)

if __name__ == "__main__":
    main()
```

StackHeapsumOfRow
(7k)

arr 7k

sum (15)
return 15main()

mat 7k

ans 15

9	5	1
0	0	0



Quizzes

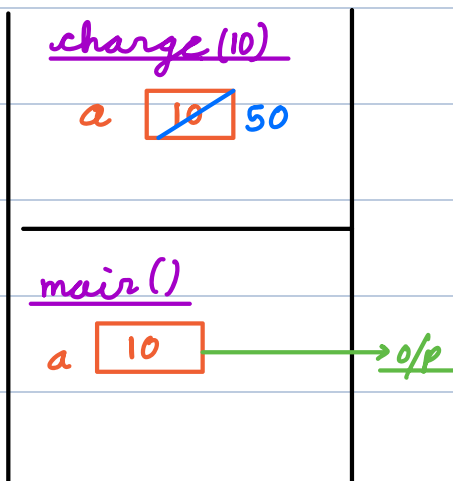
Java

```
static void change(int a) {  
    a = 50;  
}  
  
public static void main(String args[]) {  
    int a = 10;  
    change(a);  
    System.out.println(a);  
}
```

Python

```
def change(a):  
    a = 50  
  
def main():  
    a = 10  
    change(a)  
    print(a)  
  
if __name__ == "__main__":  
    main()
```

Stack



Heap



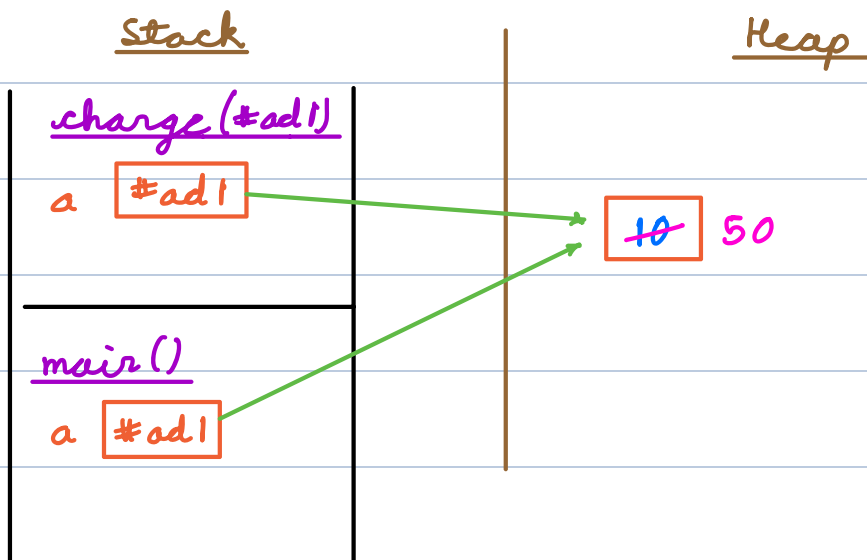
Java

```
static void change(int[] a) {  
    a[0] = 50;  
}  
  
public static void main(String args[]) {  
    int[] a = {10};  
    change(a);  
    System.out.println(a[0]);  
}
```

50

Python

```
def change(a):  
    a[0] = 50  
  
def main():  
    a = [10]  
    change(a)  
    print(a[0])  
  
if __name__ == "__main__":  
    main()
```



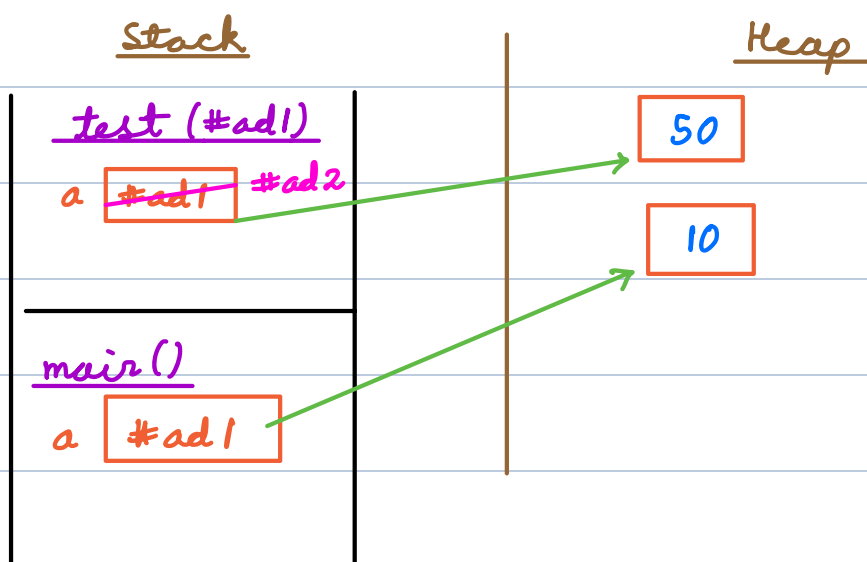


Java

```
static void test(int[] a) {  
    a = new int[1];  
    a[0] = 50;  
}  
  
public static void main(String args[]) {  
    int[] a = {10};  
    test(a);  
    System.out.println(a[0]);  
}
```

Python

```
def test(a):  
    a = [0]  
    a[0] = 50  
  
def main():  
    a = [10]  
    test(a)  
    print(a[0])  
  
if __name__ == "__main__":  
    main()
```





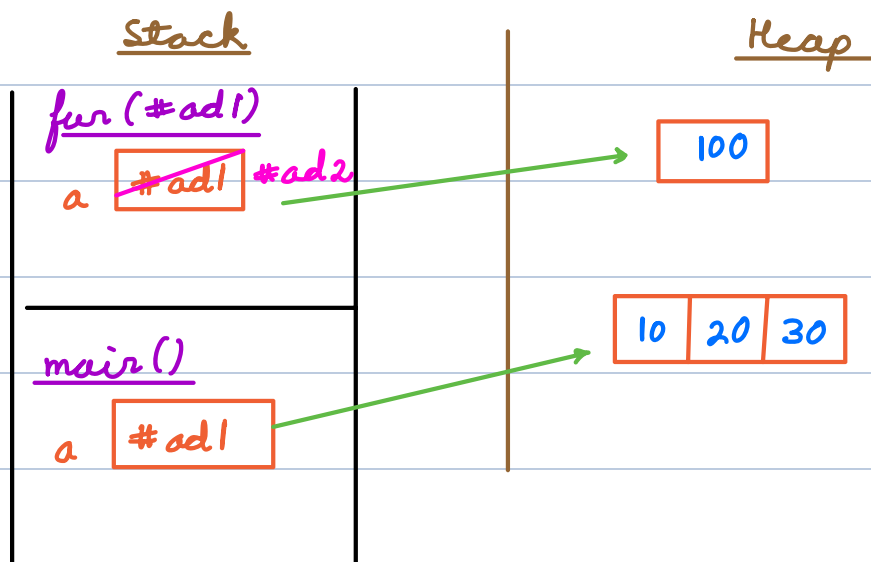
Java

```
static void fun(int[] a) {  
    a = new int[1];  
    a[0] = 100;  
}  
  
public static void main() {  
    int[] a = {10, 20, 30};  
    fun(a);  
    System.out.println(a[0]);  
}
```

10

Python

```
def fun(a):  
    a = [0]  
    a[0] = 100  
  
def main():  
    a = [10, 20, 30]  
    fun(a)  
    print(a[0])  
  
if __name__ == "__main__":  
    main()
```





Java

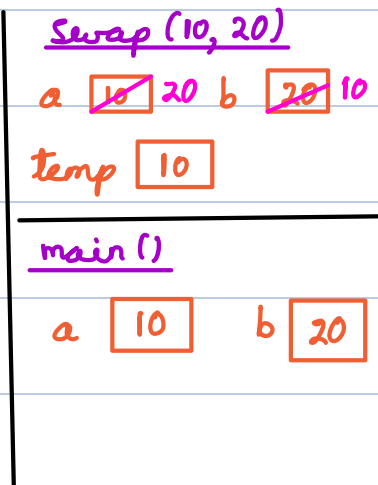
```
static void swap(int a,int b) {  
    int temp = a;  
    a = b;  
    b = temp;  
}  
  
public static void main(String args[]) {  
    int a = 10;  
    int b = 20;  
    swap(a,b);  
    System.out.println(a + " " + b);  
}
```

10 20

Python

```
def swap(a, b):  
    temp = a  
    a = b  
    b = temp  
  
def main():  
    a = 10  
    b = 20  
    swap(a, b)  
    print(a, b)  
  
if __name__ == "__main__":  
    main()
```

Stack



Heap



Java

```
static void swap(int[] a, int[] b) {  
    int temp = a[0];  
    a[0] = b[0];  
    b[0] = temp;  
}  
  
public static void main(String args[]) {  
    int[] a = {10};  
    int[] b = {20};  
    swap(a, b);  
    System.out.println(a[0] + " " + b[0]);  
}
```

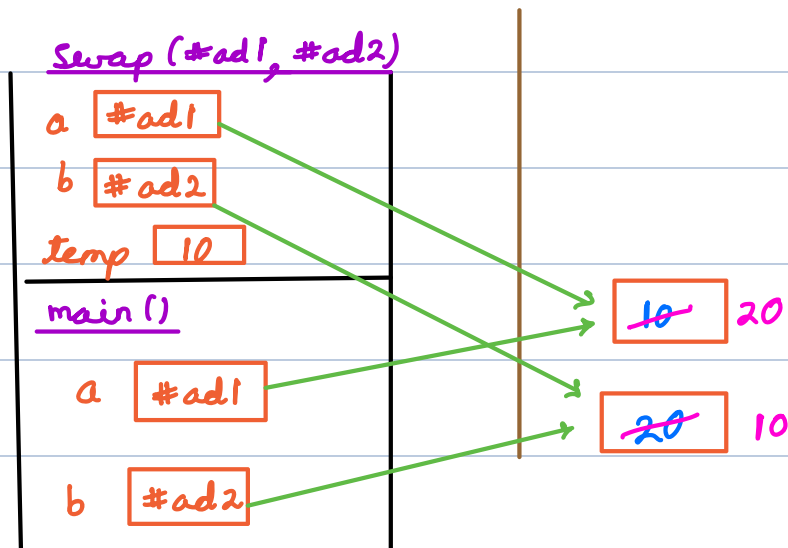
20 10

Python

```
def swap(a, b):  
    temp = a[0]  
    a[0] = b[0]  
    b[0] = temp  
  
def main():  
    a = [10]  
    b = [20]  
    swap(a, b)  
    print(a[0], b[0])  
  
if __name__ == "__main__":  
    main()
```

Stack

Heap





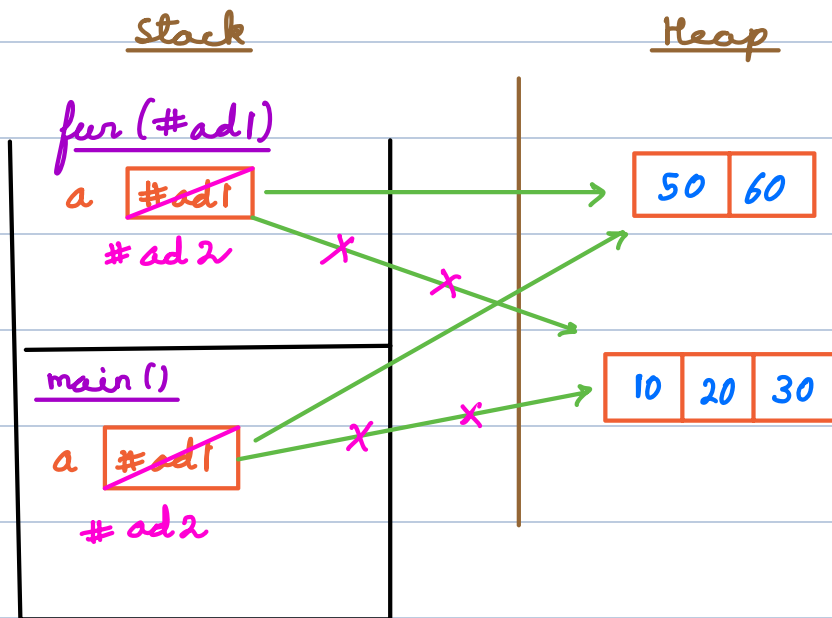
Java

```
static int[] fun(int[] a) {  
    a = new int[2];  
    a[0] = 50; a[1] = 60;  
    return a;  
}  
  
public static void main(String args[]) {  
    int[] a = {10, 20, 30};  
    a = fun(a);  
    System.out.println(a[0]);  
}
```

50

Python

```
def fun(a):  
    a = [0, 0]  
    a[0] = 50  
    a[1] = 60  
    return a  
  
def main():  
    a = [10, 20, 30]  
    a = fun(a)  
    print(a[0])  
  
if __name__ == "__main__":  
    main()
```





Java

```
static void test(int[] a) {  
    a = new int[2];  
    a[0] = 94;  
}  
  
public static void main(String args[]) {  
    int[] a = {10, 20, 30};  
    test(a);  
    System.out.println(a[0]); 10  
}
```

Python

```
def test(a):  
    a = [0, 0]  
    a[0] = 94  
  
def main():  
    a = [10, 20, 30]  
    test(a)  
    print(a[0])  
  
if __name__ == "__main__":  
    main()
```

